

LANE A – Acquisition & calibrated forward / observation model (physical, measured)

Micro-scan acquisition

Step-and-shoot raster; 248-frame clean main session.
Sub-pixel shifts from stage command via $\theta = 47.6^\circ \pm 0.1^\circ$, detector pitch $10 \mu\text{m}/\text{px}$ ($40 \mu\text{m} \Rightarrow 4.0 \text{px}$).
stage command = prior, not GT

Forward / observation operator

$$y_k = DBHS_{f_k} x + n_k$$

S_{f_k} : refined sub-pixel shift
 H : Gaussian PSF, $\sigma \in [0.2, 0.5]$ LR px (arbitrated)
 B : $10 \mu\text{m}$ detector box integration
 D : decimate $2 \times \rightarrow 1 \times$ grid
 n_k : noise floor 0.0724°C

Null space

$B \cdot H$ band attenuation + D aliasing $\Rightarrow$ a null space *invisible to any observation-side loss.*

TCForge synthetic

Scenes rendered with the *measured* forward model (SSAA coverage AA; PSF σ range; $10 \mu\text{m}$ box; noise floor; raster shift lattice). Burst pool, $K = 4$ drizzle variants.
real data has NO HR ground truth

...null space $\Rightarrow$ anchor blind

synthetic HR target

LANE B – Learned reconstruction pipeline (this paper)

Q2 – Information delivered?

$1 \times$ -stats collapse vs. hybrid drizzle injection

(baseline) $1 \times$ -stats input

5 channels on the $1 \times$ grid: aligned mean / median / coverage / variance / high-pass.
sub-pixel phase collapses before the network $\times$

Data-driven alignment

High-pass NCC init + contour refinement $\rightarrow$ refined shifts.
held-out contour Chamfer **$0.381 \rightarrow 0.134 \text{ px}$**

(ours) evidence-injecting hybrid input

8 ch @ $2 \times$ grid = [5 upsampled $1 \times$ -stats] $\oplus$ [3 drizzle@ $2 \times$: mean (**ch5**), coverage (ch6), variance (ch7)]; network runs at scale = 1 on the $2 \times$ grid.
ch5 = drizzle mean = most observation-faithful channel (added back $1 \times$)

Q1 – Information exists?

split-half FRC & MAP-TV gain

Backbone: UNet

Plain UNet (base 64), encoder-decoder, bilinear HR head.
architecture held fixed; PixelShuffle head rejected (negative result).
output: residual δ

Q3 – Information faithful?

residual remedy; loss-side anchor fails

Residual-over-observation head

$$\hat{x} = \underset{\text{obs. base}}{\text{ch5}} + \delta$$

penalty $\lambda \cdot \text{mean}|\delta|$ (L1).
 $\delta = 0 \Rightarrow$ keep the most-faithful observation by default; penalty makes hallucination costly everywhere (no null space to hide in).
 $\lambda \in \{0.2, 0.5, 1.2, 3.0\}$

Reconstr.

$\hat{x}$: **$2 \times$ contour-level** output ($5 \mu\text{m}$ sample, $10 \mu\text{m}$ Nyquist period).

248 aligned frames

ch5 drizzle mean (observation base added back)

reconstruction

LANE C – Classical anchors & GT-free evaluation (arbitration)

Classical anchors

- **drizzle** — minimal-assumption fusion
- **MAP-TV** deconvolution (FISTA, acceptance anchor)
- **anisotropic coverage-weighted TGV** — stripe-artifact fix (artifact **$3.870 \rightarrow 0.695$**).

GT-free evaluation / selection

- phase-stratified **split-half FRC** (cutoff $17.0 \mu\text{m}$)
- observation-anchored proxy pair (artifact $\downarrow$ / raw-control corr $\uparrow$)
- structural grain proxy **lattice**
- dual-domain visual gate (temperature + high-pass)
- mechanical checkpoint selection on the proxy Pareto.

Verdict (GT-free)

Complementary failure modes — TGV imprints a mild TV staircase (“beading”); evidence-injected learning is sharpest but carries the most (unverifiable) high-frequency grain.
No GT-certifiable single winner.